

# COURIER MANAGEMENT SYSTEM(CMS)

## 6. PROJECT TESTING PHASE

### 6.1 Testing Phase

**Project Testing Phase** is the process of checking whether the Courier Management System works correctly according to the specified requirements.

**Main objectives:**

- Identify errors and bugs.
- Verify each system function.
- Check data accuracy and security.
- Ensure smooth communication between modules.
- Confirm that the system meets customer and staff requirements.

### 6.2 Testing Objectives

- Verify that all modules work correctly.
- Ensure customer and shipment data is stored accurately.
- Check shipment tracking and status updates.
- Verify delivery agent assignments.
- Test invoice generation.
- Check reports and dashboards.
- Ensure secure login and access control.
- Improve system reliability and usability.

### 6.3 Functional Testing

| Module              | Testing                               |
|---------------------|---------------------------------------|
| Customer Management | Add, update and view customer details |
| Shipment Booking    | Create and manage shipments           |
| Agent Assignment    | Assign shipments to delivery agents   |
| Shipment Tracking   | Check current shipment status         |
| Notifications       | Verify delivery notifications         |

| Module             | Testing                      |
|--------------------|------------------------------|
| Invoice Generation | Generate correct invoice     |
| Reports            | Display accurate information |
| Login              | Verify authentication        |

## 6.4 Sample Test Cases

| Test Case             | Expected Result                  | Status |
|-----------------------|----------------------------------|--------|
| Customer registration | Customer registered successfully | Pass   |
| Shipment booking      | Shipment created successfully    | Pass   |
| Agent assignment      | Agent assigned successfully      | Pass   |
| Tracking ID search    | Correct status displayed         | Pass   |
| Status update         | New status saved correctly       | Pass   |
| Invoice generation    | Correct invoice generated        | Pass   |
| Login                 | Valid user can access system     | Pass   |
| Reports               | Accurate data displayed          | Pass   |

## 6.5 Non-Functional Testing

### Performance Testing

- Checks system response time.
- Ensures multiple requests can be handled.

### Security Testing

- Tests login and authentication.
- Protects customer and shipment information.

### Usability Testing

- Checks whether the interface is simple and user-friendly.

### Reliability Testing

- Ensures the system works consistently without frequent errors.

### **Compatibility Testing**

- Checks the system on different browsers and devices.

## **6.6 Error Identification and Correction**

During testing, errors are identified and corrected.

### **Common errors checked:**

- Invalid login details.
- Missing customer information.
- Incorrect shipment details.
- Invalid tracking ID.
- Incorrect invoice calculation.
- Wrong delivery status.
- Missing notifications.
- Incorrect report data.

After fixing an error, the feature is tested again to confirm that the problem has been resolved.

## **6.7 Testing Result**

After completing the testing process:

- Major system functions work correctly.
- Shipment information is properly maintained.
- Tracking and status updates work correctly.
- Delivery agent assignments are verified.
- Invoices are generated accurately.
- Reports and dashboards display required information.
- Security and authentication are checked.
- The system is ready for demonstration and deployment after successful testing.